Practice Set – Unit I

Sets, Subsets, Set Operations, Counting, Cartesian Product, and Relations

MAT113 – Mathematics for Biosciences

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Dr. Lokesh Singh
Department of Mathematics,
Amity University Punjab, Mohali.
lsingh@pb.amity.edu

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Instructions

- Attempt all questions in your notebook.
- Show all steps clearly.
- Venn diagram questions must include the diagram with shaded regions.

1 Section A: Set Notation and Types (5 Questions)

A1. Write the following in **roster form**:

(a) $\{x \mid x \text{ is a vowel in English}\}$

(b) $\{x \mid x \text{ is a day starting with T}\}$

(c) $\{x \in \mathbb{N} \mid x < 8,\ x \text{ is odd}\}$

A2. Write the following in **set-builder form**:

(a) $\{2, 4, 6, 8, 10\}$

(b) $\{\text{gold, silver, bronze}\}$

(c) $\{1, 4, 9, 16, 25\}$

A3. Find the **cardinality** of each set:

(a) $A = \{\text{apple, banana, mango, orange, grapes}\}$

(b) $B = \{2, 4, 6, 8, 10, 12, 14\}$

(c) $C = \varnothing$

A4. State whether each is **True** or **False**:

(a) $\{1, 2, 3\} \subset \{1, 2, 3, 4, 5\}$

(b) $\{1, 2, 3\} \subseteq \{1, 2, 3\}$

(c) $\varnothing \subset \{a\}$

(d) $\{a, b\} \subset \{a, b\}$

A5. Let $A = \{a, b\}$. Write the **power set** $\mathcal{P}(A)$. Also find $n(\mathcal{P}(A))$.

2 Section B: Union, Intersection, Difference (17 Questions)

B1. $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7, 8\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.

B2. $A = \{\text{apple, banana, mango}\}$, $B = \{\text{mango, orange, grapes}\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.

B3. $A = \{\text{cricket, football, hockey}\}$, $B = \{\text{tennis, badminton, cricket}\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.

B4. $A = \{2, 4, 6, 8, 10\}$, $B = \{3, 6, 9\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.

B5. $A = \{a, e, i, o, u\}$, $B = \{a, b, c, d, e\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.

B6. $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$. Are A and B disjoint?

B7. $A = \varnothing$, $B = \{1, 2, 3\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.

B8. $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6\}$. Find $A \cap B$, $A - B$, and verify $A - B = B'$ within A.

B9. $A = \{x \mid x \text{ is even},\ x < 12\}$, $B = \{x \mid x \text{ is prime},\ x < 12\}$. Write both sets in roster form, then find $A \cup B$, $A \cap B$, $A - B$.

B10. $A = \{\text{red, blue, green}\}$, $B = \{\text{yellow, orange, purple}\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.

B11. $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$. Find A', B', $(A \cup B)'$, $(A \cap B)'$.

B12. $A = \{\text{mango, apple}\}$, $B = \{\text{banana, grapes}\}$, $C = \{\text{apple, banana}\}$. Find $A \cup B \cup C$, $A \cap B \cap C$, $A \cap C$, $B \cup C$.

B13. With $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$. **Verify De Morgan's Law 1:** $(A \cup B)' = A' \cap B'$.

B14. With the same sets as B13, **verify De Morgan's Law 2:** $(A \cap B)' = A' \cup B'$.

B15. $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4, 6, 8\}$. Find $A \cap B$, $A - B$, $B - A$, $A \cup B$. Also find $n(A \cup B)$ and verify with inclusion-exclusion.

B16. $A = \{x \mid x \in \mathbb{N},\ x \leq 10\}$, $B = \{x \mid x \in \mathbb{N},\ x \leq 10,\ x \text{ is even}\}$. Find B' (within A as universal set). How many odd numbers are there?

B17. $A = \{\text{Jan, Feb, Mar}\}$, $B = \{\text{Mar, Apr, May}\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.

B18. Given $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8\}$, $C = \{3, 6, 9\}$. Find:

(a) $A \cap B \cap C$

(b) $(A \cup B) \cap C$

(c) $A - (B \cup C)$

(d) $(A - B) \cup C$

3 Section C: Venn Diagrams (10 Questions)

For each question, draw a Venn diagram on your notebook and shade the requested region.

C1. $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$. Draw a Venn diagram and shade $A \cup B$.

C2. Using the same sets as C1, shade $A \cap B$.

C3. Using the same sets as C1, shade $A - B$.

C4. Using the same sets as C1, shade $B - A$.

C5. $A = \{1, 2, 3, 4, 5\}$. Draw a Venn diagram with universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and shade A'.

C6. Draw a Venn diagram for $A \subset B$, where $A = \{1, 2\}$ and $B = \{1, 2, 3, 4\}$.

C7. Draw a Venn diagram showing two **disjoint sets** $A = \{1, 2\}$ and $B = \{3, 4\}$ with $U = \{1, 2, 3, 4, 5, 6\}$.

C8. Draw a Venn diagram and shade $(A \cup B)'$.

C9. Draw a 3-set Venn diagram with sets A, B, C and shade $A \cap B \cap C$.

C10. Draw a 3-set Venn diagram and shade $(A \cup B) - C$.

4 Section D: Counting of Sets (12 Questions)

D1. If $n(A) = 15$, $n(B) = 20$, $n(A \cap B) = 8$, find $n(A \cup B)$.

D2. If $n(A) = 25$, $n(B) = 30$, $n(A \cup B) = 45$, find $n(A \cap B)$.

D3. If $n(U) = 50$, $n(A) = 32$, find $n(A')$.

D4. If $n(A) = 20$, $n(A \cap B) = 7$, find $n(A - B)$.

D5. If $n(A \times B) = 24$ and $n(A) = 6$, find $n(B)$.

D6. If $n(A) = 3$, find $n(\mathcal{P}(A))$ and the number of **proper** subsets of A.

D7. In a survey of 60 people, 35 like tea, 25 like coffee, 15 like both.

(a) How many like at least one beverage?

(b) How many like neither?

(c) How many like only tea?

D8. In a class of 40 students, 25 play cricket, 20 play football, 10 play both.

(a) How many play at least one sport?

(b) How many play only cricket?

(c) How many play only football?

(d) How many play neither?

D9. In a group of 100 students, 60 study Mathematics, 45 study Physics, 25 study both.

(a) How many study at least one subject?

(b) How many study neither?

D10. If $n(A) = 40$, $n(B) = 35$, $n(A \cup B) = 60$, find $n(A \cap B)$.

D11. If $n(A) = 5$, $n(B) = 7$, $n(C) = 3$, find $n(A \times B \times C)$.

D12. If $n(A) = 4$, how many subsets does A have? How many **proper** subsets?

5 Section E: Inclusion-Exclusion (5 Questions)

E1. In a group of 50 people, 30 read Hindi newspaper, 25 read English newspaper, 10 read both.

(a) How many read at least one newspaper?

(b) How many read exactly one?

(c) How many read neither?

E2. A survey of 200 people shows: 120 use WhatsApp, 90 use Instagram, 60 use both.

(a) How many use at least one?

(b) How many use neither?

E3. 150 students: 90 like pizza, 70 like burger, 50 like both.

(a) How many like at least one?

(b) How many like only pizza?

(c) How many like neither?

E4. In a club of 40 members, 25 speak Hindi, 20 speak English, 15 speak both.

(a) How many speak at least one language?

(b) How many speak only Hindi?

(c) How many speak neither?

E5. In a survey, $n(A \cup B) = 45$, $n(A) = 30$, $n(B) = 25$. Find $n(A \cap B)$. Verify your answer using a Venn diagram.

6 Section F: Cartesian Product (8 Questions)

F1. $A = \{1, 2\}$, $B = \{a, b, c\}$. Find $A \times B$ and $B \times A$. Are they equal?

F2. Find $n(A \times B)$ if $n(A) = 4$ and $n(B) = 5$.

F3. $A = \{1, 2, 3\}$, $B = \{x, y\}$. Write $A \times B$ in roster form.

F4. Write all ordered pairs in $\{a, b\} \times \{1, 2\}$.

F5. If $(2x - 1,\ y + 3) = (5, 7)$, find x and y.

F6. $A = \{1, 2\}$, $B = \{3, 4\}$, $C = \{5\}$. Find $n(A \times B \times C)$.

F7. Is $(3, 5) = (5, 3)$? Explain with a reason.

F8. $A = \{\text{cricket, football}\}$, $B = \{\text{India, Australia}\}$. Write $A \times B$ and $B \times A$.

7 Section G: Relations (7 Questions)

G1. $A = \{1, 2, 3\}$, $B = \{2, 4, 6\}$. Define R by $b = 2a$. Write R in roster form.

G2. Find the **domain** and **range** of $R = \{(1, 2),\ (2, 4),\ (3, 6)\}$.

G3. On $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$.

(a) Is R reflexive?

(b) Is R symmetric?

(c) Is R transitive?

G4. On $A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 1), (1, 3), (3, 1)\}$. Is R symmetric? Justify.

G5. $A = \{1, 2, 3, 4\}$, $R = \{(a, b) \mid a + b = 5\}$.

(a) Write R in roster form.

(b) Find the domain and range of R.

G6. Is $R = \{(1, 1), (2, 2), (3, 3)\}$ an equivalence relation on $\{1, 2, 3\}$? Justify.

G7. On $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$. Is R an equivalence relation? Justify.

8 Section H: Relation is a Function or Not? (6 Questions)

H1. $A = \{1, 2, 3\}$, $B = \{a, b, c\}$, $R = \{(1, a), (2, b), (3, c)\}$. Is R a function? Explain.

H2. $A = \{1, 2, 3\}$, $B = \{a, b, c\}$, $R = \{(1, a), (1, b), (2, c), (3, a)\}$. Is R a function? Explain.

H3. $A = \{1, 2, 3\}$, $B = \{a, b, c\}$, $R = \{(1, a), (2, b)\}$. Is R a function? Explain.

H4. $A = \{1, 2, 3\}$, $B = \{a, b, c\}$, $R = \{(1, a), (2, a), (3, a)\}$. Is R a function? Explain.

H5. $A = \{1, 2, 3, 4\}$, $R = \{(1, 1), (2, 4), (3, 9), (4, 16)\}$. Is R a function? Explain.

H6. $A = \{1, 2, 3\}$, $R = \{(1, 2), (1, 3), (2, 4)\}$. Is R a function? Explain.

9 Section I: Mixed and Challenge Problems (5 Questions)

I1. In a class of 50 students:

- 30 passed Maths
- 25 passed Physics
- 15 passed both

(a) Draw a labelled Venn diagram.

(b) Find the number who passed only Maths.

(c) Find the number who passed only Physics.

(d) Find the number who passed neither.

I2. In a survey of 500 people:

- 300 use WhatsApp
- 250 use Facebook
- 100 use both

(a) Find the number who use at least one platform.

(b) Find the number who use only WhatsApp.

(c) Find the number who use neither.

(d) Verify your answer using the inclusion-exclusion formula.

I3. $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8\}$, $C = \{3, 6, 9\}$. Find:

(a) $A \cap B \cap C$

(b) $(A \cup B) \cap C$

(c) $A - (B \cup C)$

(d) $(A - B) \cup C$

I4. Let $A = \{a, b, c\}$. Write $\mathcal{P}(A)$. Find $n(\mathcal{P}(A))$ and the number of proper subsets.

I5. In a hostel of 100 students:

- 60 drink tea
- 40 drink coffee
- 20 drink both

(a) How many drink neither?

(b) How many drink exactly one?

(c) Draw a fully labelled Venn diagram.